

Nathanael Senjaya

📍 Seoul, South Korea

🇮🇩 Indonesia

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in nathanael senjaya

Education

- March 2026 ~  M.S. Korea University, South Korea
in Department of Artificial Intelligence
◦ Advisor : **Prof. Chanwoo Kim**
- January 2024  B.S. National Taiwan Normal University, Taiwan
in Department of Electrical Engineering
Project : "Facial-Based Anxiety Detection from Video Using Machine Learning"
◦ Developed ensemble machine learning models (SVM, KNN, Random Forest) and analysis for anxiety detection using facial feature data.
◦ Advisor : **Prof. Cheng-Kai Lu**
- August 2019  Graduated from SMAK 1 BPK PENABUR Bandung, Indonesia in **Natural Science Majors**

Research Interest

Emotion-Centered AI, Speech Processing, Large Language Model (LLM), Natural Language Processing (NLP), Machine Learning (ML), Information Retrieval (IR)

Experience

- **Volunteer Researcher**
Research Center for Information Technology Innovation (CITI), Academia Sinica Taiwan
Project : "Fine-Tuning Large Language Models for Verilog Code Generation" February 2024 - February 2025
 - Fine-tuned various open-source Large Language Models (LLMs) both original and QLoRA for Verilog code generation, achieving competitive result to state-of-the-art.
 - Advisor : **Prof. Chuan-Ju Wang**
- **Summer Internship**
Research Center for Information Technology Innovation (CITI), Academia Sinica Taiwan
Project : "Document Retrieval via Key-points Augmentation and LLM-Based Question Generation" July 2024 - August 2024
 - Developed a novel query expansion technique leveraging LLM-generated questions derived from context-enriched key points, enhancing information retrieval performance in knowledge-intensive tasks.
 - Advisor : **Prof. Chuan-Ju Wang**
- **Participant** of 2023 Statistics Camp (統計科學營) hosted by *Academia Sinica Institute of Statistics, Taiwan* (23 - 24 August 2023).
- **Participant** of numerous online courses from different well-known platforms: *Machine Learning, Python, SQL, Data Visualization, Git, Java.*

Skills

- Programming Languages: Python, C , R, etc.
- Libraries & Frameworks: TensorFlow, PyTorch, HuggingFace, LangChain, Transformers, spaCy, SentenceTransformers, Scikit-learn, Pandas, NumPy, etc.
- Data Handling: MongoDB.
- Tools & Environments: Jupyter Notebooks, Linux, Git

Language

1. English : Academic IELTS Band 7.5 (Proficient)
2. Mandarin : Conversational-level